

Examples

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Example 1

On any given day, a device can be either **on** or **off**.

- ▶ If the device is on today, the probability that it will be off tomorrow is $1/4$.
- ▶ If the device is off today, the probability that it will remain off tomorrow is $1/6$.
- ▶ If the device remains off for three consecutive days, it is replaced by a new device.

Example 1

Let X_n represent the state of the device on day n . The state is defined as 0 if the device is on. If the device has been off for i consecutive days, the state is i , where $i = 1, 2, 3$.

The transition probability matrix is:

$$P = \begin{pmatrix} 3/4 & 1/4 & 0 & 0 \\ 5/6 & 0 & 1/6 & 0 \\ 5/6 & 0 & 0 & 1/6 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

- The Markov chain is **irreducible** and **aperiodic**.

Example 1

For instance, let us calculate the probability that the device will remain off for the next three consecutive days, given that it is operational today.

$$\begin{aligned} & \mathbb{P}(X_3 = 3, X_2 = 2, X_1 = 1 \mid X_0 = 0) \\ &= \mathbb{P}(X_3 = 3 \mid X_2 = 2, X_1 = 1, X_0 = 0) \mathbb{P}(X_2 = 2, X_1 = 1 \mid X_0 = 0) \\ &= \mathbb{P}(X_3 = 3 \mid X_2 = 2, X_1 = 1, X_0 = 0) \mathbb{P}(X_2 = 2 \mid X_1 = 1, X_0 = 0) \\ &\quad \mathbb{P}(X_1 = 1 \mid X_0 = 0) \\ &= p_{23} p_{12} p_{01} = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{4} = \frac{1}{144} \end{aligned}$$

Example 1

Now, let us calculate the stationary distribution. The balance equation are:

$$\left\{ \begin{array}{lcl} \frac{1}{4}\pi_0 & = & \frac{5}{6}\pi_1 + \frac{5}{6}\pi_2 + \pi_3 \\ \pi_1 & = & \frac{1}{4}\pi_0 \\ \pi_2 & = & \frac{1}{6}\pi_1 \\ \pi_3 & = & \frac{1}{6}\pi_2 \end{array} \right.$$

The solution that satisfies $\sum_{i=0}^3 \pi_i = 1$ is:

$$\pi_0 = \frac{144}{187}, \quad \pi_1 = \frac{36}{187}, \quad \pi_2 = \frac{6}{187}, \quad \pi_3 = \frac{1}{187}$$

Example 1

- ▶ The probability π_i represents the inverse of the mean recurrence time μ_i , indicating the long-term expected fraction of time that the chain spends in state i .
- ▶ For instance, if the device remains inactive for three days, it is replaced with a new unit. The expected duration until this replacement device also requires substitution is given by

$$\mu_3 = \frac{1}{\pi_3} = 187 \text{ days}$$

Example 1

Finally, suppose now that if the device is off for two consecutive days, it is removed and not replaced.

This new situation can be modelled by an absorbing Markov chain with transition probability matrix

$$T = \begin{pmatrix} 3/4 & 1/4 & 0 \\ 5/6 & 0 & 1/6 \\ 0 & 0 & 1 \end{pmatrix}$$

- State 2 is the unique absorbing state.

Example 1

The fundamental matrix is

$$\begin{aligned} \mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1} &= \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 3/4 & 1/4 \\ 5/6 & 0 \end{pmatrix} \right)^{-1} \\ &= \begin{pmatrix} 1/4 & -1/4 \\ -5/6 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 24 & 6 \\ 20 & 6 \end{pmatrix} \end{aligned}$$

For example, if the device is on, then the expected number of days until it will be removed is $24 + 6 = 30$.

Example 2

A bus service operating between cities A and B provides three wheelchairs for passenger use.

- ▶ For each journey between the cities, one of the following scenarios occurs:
 - A single request for a wheelchair occurs with probability p , where $0 < p < 1$.
 - No request occurs with probability $q = 1 - p$.
- ▶ If a request for a wheelchair is made at the start of a journey, one available wheelchair (if any remain) is used and left at the journey's destination.

Example 2

Let X_k be the number of wheelchairs that are available at the initial endpoint of the k -th journey between the cities.

Then X_k takes values on $S = \{0, 1, 2, 3\}$, and the transition probabilities are:

$$\mathbb{P}(X_{k+1} = 3 \mid X_k = 0) = 1$$

$$\mathbb{P}(X_{k+1} = 3 \mid X_k = 1) = p, \quad \mathbb{P}(X_{k+1} = 2 \mid X_k = 1) = q$$

$$\mathbb{P}(X_{k+1} = 2 \mid X_k = 2) = p, \quad \mathbb{P}(X_{k+1} = 1 \mid X_k = 2) = q$$

$$\mathbb{P}(X_{k+1} = 1 \mid X_k = 3) = p, \quad \mathbb{P}(X_{k+1} = 0 \mid X_k = 3) = q$$

Example 2

Therefore, we have a Markov chain such that

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & q & p \\ 0 & q & p & 0 \\ q & p & 0 & 0 \end{pmatrix}$$

- ▶ The chain is irreducible.
- ▶ It is also aperiodic (there is a loop at state 2).

Example 2

Assume the bus commences its k -th journey at location B with i available wheelchairs, where $0 \leq j \leq 3$.

The probability distribution for the number of wheelchairs available upon the bus's return to B is given by

$$\mathbb{P}(X_{k+2} = j \mid X_k = i), \quad 0 \leq j \leq 3$$

Example 2

These probabilities can be obtained from $\mathbf{P}^2$:

$$\mathbf{P}^2 = \begin{pmatrix} q & p & 0 & 0 \\ pq & p^2 + q^2 & pq & 0 \\ 0 & pq & p^2 + q^2 & pq \\ 0 & 0 & pq & q + p^2 \end{pmatrix}$$

- ▶ The $i + 1$ -th row, $0 \leq i \leq 3$, corresponds to the probability distribution of the variable X_{k+2} , given that the bus started from B with i available chairs.

Example 2

The balance equations are

$$\left\{ \begin{array}{lcl} \pi_0 & = & q \pi_3 \\ \pi_1 & = & q \pi_2 + p \pi_3 \\ q \pi_2 & = & q \pi_1 \\ \pi_3 & = & \pi_0 + p \pi_1 \end{array} \right.$$

The solution satisfying $\sum_{i=0}^3 \pi_i = 1$ is

$$\pi = \left(\frac{q}{3+q}, \frac{1}{3+q}, \frac{1}{3+q}, \frac{1}{3+q} \right)$$

Example 2

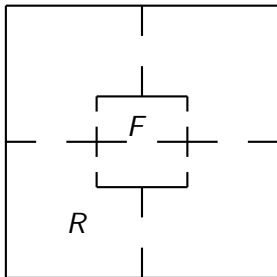
$$\pi = \left(\frac{q}{3+q}, \frac{1}{3+q}, \frac{1}{3+q}, \frac{1}{3+q} \right)$$

- ▶ For instance, the probability that a request cannot be fulfilled due to all wheelchairs being located at the opposite endpoint is given by $p\pi_0 = pq/(3+q)$.
- ▶ Assuming that all three wheelchairs are initially available at the starting point, the expected number of journeys between the cities required for all three wheelchairs to be available again at an endpoint is

$$\mu_3 = \frac{1}{\pi_3} = 3 + q$$

Example 3

A mouse is in the maze of the figure. It randomly selects (with equal probability) one of the doors in each room and enters the corresponding adjacent room.



Example 3

Labelling the rooms from top to down and from left to right we have the following transition probability matrix:

$$\mathbf{P} = \begin{pmatrix} 0 & 1/3 & 1/3 & 1/3 & 0 & 0 \\ 1/3 & 0 & 1/3 & 0 & 0 & 1/3 \\ 1/3 & 1/3 & 0 & 0 & 1/3 & 0 \\ 1/3 & 0 & 0 & 0 & 1/3 & 1/3 \\ 0 & 0 & 1/3 & 1/3 & 0 & 1/3 \\ 0 & 1/3 & 0 & 1/3 & 1/3 & 0 \end{pmatrix}$$

Example 3

If the mouse starts in the top-left room, the probability distribution after two movements, $\mathbf{p}^{(2)}$, is given by:

$$\begin{aligned}\mathbf{p}^{(2)} &= (1, 0, 0, 0, 0, 0) \cdot \mathbf{P}^2 \\ &= (1, 0, 0, 0, 0, 0) \cdot \begin{pmatrix} \frac{1}{3} & \frac{1}{9} & \frac{1}{9} & 0 & \frac{2}{9} & \frac{2}{9} \\ \frac{1}{9} & \frac{1}{3} & \frac{1}{9} & \frac{2}{9} & \frac{2}{9} & 0 \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{3} & \frac{2}{9} & 0 & \frac{2}{9} \\ 0 & \frac{2}{9} & \frac{2}{9} & \frac{1}{3} & \frac{1}{9} & \frac{1}{9} \\ \frac{2}{9} & \frac{2}{9} & 0 & \frac{1}{9} & \frac{1}{3} & \frac{1}{9} \\ \frac{2}{9} & 0 & \frac{2}{9} & \frac{1}{9} & \frac{1}{9} & \frac{1}{3} \end{pmatrix} \\ &= \left(\frac{1}{3}, \frac{1}{9}, \frac{1}{9}, 0, \frac{2}{9}, \frac{2}{9} \right)\end{aligned}$$

Example 3

The chain is **irreducible**. Moreover, notice that $\mathbf{P}$ is doubly stochastic. Therefore, the stationary distribution is **uniform**:

$$\pi = \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6} \right)$$

The chain is also **aperiodic** (why?). Hence, for any i, j we have that

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = j \mid X_0 = i) = \frac{1}{6}$$

Example 3

Finally, suppose that if the mouse enters room F , it will stay there forever due to the abundance of cheese. However, if it visits room R , it will get caught in a mousetrap.

We must consider an absorbing Markov chain that has absorbing states F and R . By reordering the states, we can express them as $S = \{1, 2, 5, 6, F = 3, R = 4\}$.

$$P = \left(\begin{array}{cccc|cc} 0 & 1/3 & 0 & 0 & 1/3 & 1/3 \\ 1/3 & 0 & 0 & 1/3 & 1/3 & 0 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 1/3 & 1/3 & 0 & 0 & 1/3 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right)$$

Example 3

Suppose that we want to find the probability that the mouse gets caught in the mousetrap if it starts in room 1.

Let

$$Q = \begin{pmatrix} 0 & 1/3 & 0 & 0 \\ 1/3 & 0 & 0 & 1/3 \\ 0 & 0 & 0 & 1/3 \\ 0 & 1/3 & 1/3 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 1/3 & 1/3 \\ 1/3 & 0 \\ 1/3 & 1/3 \\ 0 & 1/3 \end{pmatrix}$$

Example 3

The absorption probabilities are

$$\begin{aligned} \mathbf{B} &= (\mathbf{I} - \mathbf{Q})^{-1} \mathbf{R} \\ &= \frac{1}{55} \begin{pmatrix} 63 & 24 & 3 & 9 \\ 24 & 72 & 9 & 27 \\ 3 & 9 & 63 & 24 \\ 9 & 27 & 24 & 72 \end{pmatrix} \cdot \begin{pmatrix} 1/3 & 1/3 \\ 1/3 & 0 \\ 1/3 & 1/3 \\ 0 & 1/3 \end{pmatrix} \\ &= \begin{pmatrix} 6/11 & 5/11 \\ 7/11 & 4/11 \\ 5/11 & 6/11 \\ 4/11 & 7/11 \end{pmatrix} \end{aligned}$$

► Hence the probability we are looking for is $(\mathbf{B})_{12} = 5/11$.